HEALTHCARE ACTUARIAL RISK INTELLIGENCE

# MEDIRISK

## Healthcare Cost Actuarial Risk Prediction and Decision Engine

### Exploratory Data Mining, Supervised Ensemble Modeling, and Interactive Full-Stack Underwriting Deployment

*Applied Machine Learning, Cross-Validation Benchmarking, and Counterfactual Decision Analysis on 2,772 Medical Insurance Cohorts*

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# List of Abbreviations

| Abbreviation | Full Definition / Terminology |
|---|---|
| **ACA** | Affordable Care Act (United States Federal Healthcare Legislation) |
| **API** | Application Programming Interface |
| **AV** | Actuarial Value |
| **BMI** | Body Mass Index ($kg/m^2$) |
| **CART** | Classification and Regression Trees |
| **CI** | Confidence Interval / Empirical Ensemble Prediction Band |
| **EDA** | Exploratory Data Analysis |
| **GLM** | Generalized Linear Model |
| **IQR** | Interquartile Range |
| **KDE** | Kernel Density Estimation |
| **MAE** | Mean Absolute Error |
| **ML** | Machine Learning |
| **OLS** | Ordinary Least Squares Regression |
| **OOP** | Out-Of-Pocket Maximum |
| **REST** | Representational State Transfer |
| **RF** | Random Forest Regressor Ensemble |
| **RMSE** | Root Mean Squared Error |
| $R^2$ | Coefficient of Determination |
| **SHAP** | SHapley Additive exPlanations |
| **UI** | User Interface |

# Chapter 1

# Project Overview and Problem Formulation

## 1.1 Context and Actuarial Background

Health insurance pricing relies on precise actuarial estimations of policyholder loss costs. Historically, insurers utilized manual rating manuals and generalized linear models (GLMs) that often struggled to capture complex, non-linear biometric interactions, particularly the compounding risk between elevated body mass index (BMI) and chronic tobacco consumption.

## 1.2 Healthcare Cost Dynamics: Skewness and Catastrophic Risk

Empirical healthcare expenditure data exhibit severe right-skewness and heteroscedasticity. While the median policyholder incurs moderate annual expenses, a small fraction of individuals account for disproportionately large claim totals due to acute hospitalizations, catastrophic conditions, and chronic diseases.

Key distributional challenges addressed in this research include:

1. **High Positive Skewness:** Raw medical charges exhibit a skewness coefficient of $+1.51$, requiring transformation to stabilize model estimation.
2. **Residual Heteroscedasticity:** Expenditure variance increases substantially alongside expected claims.
3. **Super-Additive Interactions:** The financial impact of smoking is magnified significantly among individuals classified as clinically obese.

## 1.3 Project Scope: From Data Mining to Operational Software

This project bridges the gap between theoretical data science and enterprise decision-making through a two-phase architecture:

1. **Phase 1 (Machine Learning Engineering):** Executed in `ML project 2.ipynb`, focusing on rigorous data profiling, monotonic log-transformations, feature engineering, and cross-validated benchmarking across multiple regression paradigms.
2. **Phase 2 (Full-Stack Operational Deployment):** Implemented in `app.py` and `app.html`, delivering an interactive decision-support system featuring real-time quote generation, empirical prediction intervals, automated Affordable Care Act (ACA) metallic plan tier matching, explainable feature attribution, dual profile counterfactual comparisons, and 10-year compounding trajectory simulations.

## 1.4 High-Level End-to-End Workflow

Figure 1.1 summarizes the end-to-end operational architecture of the MediRisk system, tracing the transformation of raw policyholder attributes into validated actuarial insights.

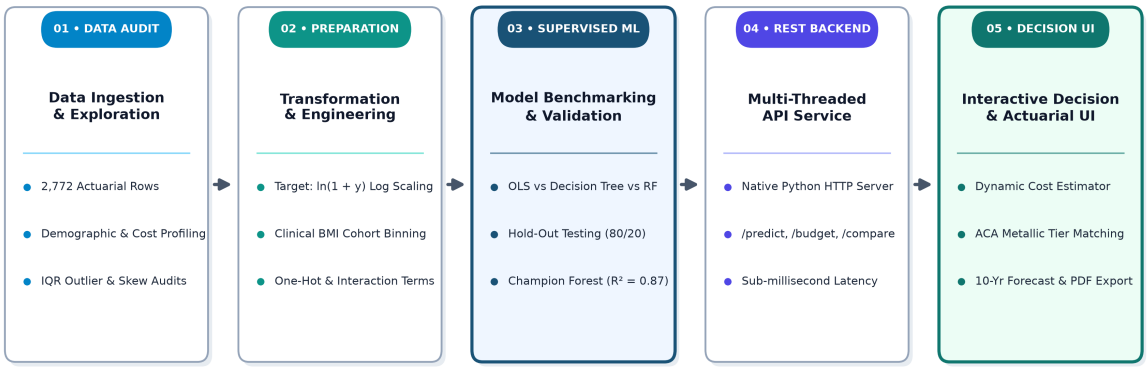



Figure 1.1: MediRisk End-to-End Architectural Pipeline

The software architecture integrates the following components:

- **Data Pipeline:** Clean feature transformation, one-hot encoding, and monotonic logarithmic scaling.
- **Serialized Inference Engine:** Pre-trained Random Forest ensemble (`best_model.joblib`) loaded in memory for instantaneous sub-millisecond predictions.
- **Multi-Threaded REST Service:** Native Python HTTP server managing request routing, JSON serialization, and concurrent access.

- **Interactive Client Interface:** Responsive single-page interface providing dynamic visual calculators and sensitivity tools.

# Chapter 2

# Exploratory Data Analysis and Cohort Characterization

## 2.1 Dataset Ingestion and Structural Auditing

The study utilizes the standardized `medical_insurance.csv` dataset comprising $N = 2,772$ observations across seven actuarial variables, with zero missing entries. Table 2.1 summarizes the dataset attributes.

Table 2.1: Summary of Actuarial Dataset Variables

| Variable | Type | Observed Range / Values | Actuarial Interpretation |
|---|---|---|---|
| `age` | Numeric | 18 to 64 years ($\mu = 39.1$) | Chronological age; baseline mortality proxy |
| `sex` | Categorical | Male (50.5%), Female (49.5%) | Biological sex indicator |
| `bmi` | Numeric | 15.96 to 53.13 ($\mu = 30.66$) | Anthropometric obesity measure |
| `children` | Integer | 0 to 5 dependents ($\mu = 1.09$) | Covered family dependency ratio |
| `smoker` | Categorical | Yes (20.5%), No (79.5%) | Primary behavioral morbidity risk factor |
| `region` | Categorical | NE, NW, SE, SW | Geographic health cost index |
| `charges` | Continuous | \$1,121.87 to \$63,770.43 | Annualized medical insurance claims ($y$) |

## 2.2 Data Integrity and Missing Value Auditing

An initial audit of all 2,772 rows confirmed complete data integrity: zero null, NaN, or corrupted values across all columns. The cohort presents a balanced gender distribution (50.5% male vs. 49.5% female), uniform geographic dispersion across four U.S. quadrants (Northeast, Northwest, Southeast, Southwest), and a 20.5% active tobacco smoking prevalence.

## 2.3 Univariate Distributions: Demographic and Financial Profiles

Table 2.2 presents the descriptive statistical metrics for the continuous variables in the cohort.

Table 2.2: Descriptive Statistics of Numerical Biometric and Financial Features

| Feature | Mean | Std. Dev. | Min | Median | Max | Skewness |
|---|---|---|---|---|---|---|
| `age` | 39.11 | 14.05 | 18.00 | 39.00 | 64.00 | +0.05 |
| `bmi` | 30.66 | 6.10 | 15.96 | 30.40 | 53.13 | +0.28 |
| `children` | 1.09 | 1.21 | 0.00 | 1.00 | 5.00 | +0.94 |
| `charges` | \$13,261.37 | \$12,111.75 | \$1,121.87 | \$9,333.01 | \$63,770.43 | +1.51 |

## 2.4 Bivariate Associations and Contingency Analysis

Correlation analysis identified chronological age ($r = +0.30$) and BMI ($r = +0.20$) as significant continuous correlates of medical charges. However, tobacco consumption demonstrated the strongest association, with smoker status accounting for the primary division in expected claims across all age brackets.

## 2.5 Outlier Diagnostics via IQR and Z-Score Boundaries

Using standard Tukey interquartile boundaries (IQR $= Q_3 - Q_1$), charges above $\$34,489.36$ qualify as mathematical outliers. However, clinical and actuarial inspection confirms these observations represent legitimate high-cost claims driven by acute care needs rather than data entry anomalies. Retaining these records is essential for preserving the tail risk required in solvency modeling.

## 2.6 The Multiplicative Synergy: BMI $\times$ Tobacco Interaction

A critical empirical finding is the non-linear interaction between elevated BMI and smoking status. As shown in Figure 2.1, smoking increases average annual claims by approximately $\$9,000$ to $\$10,000$ for non-obese individuals (BMI $< 30$), but by more than $\$30,000$ when combined with clinical obesity (BMI $\geq 30$).

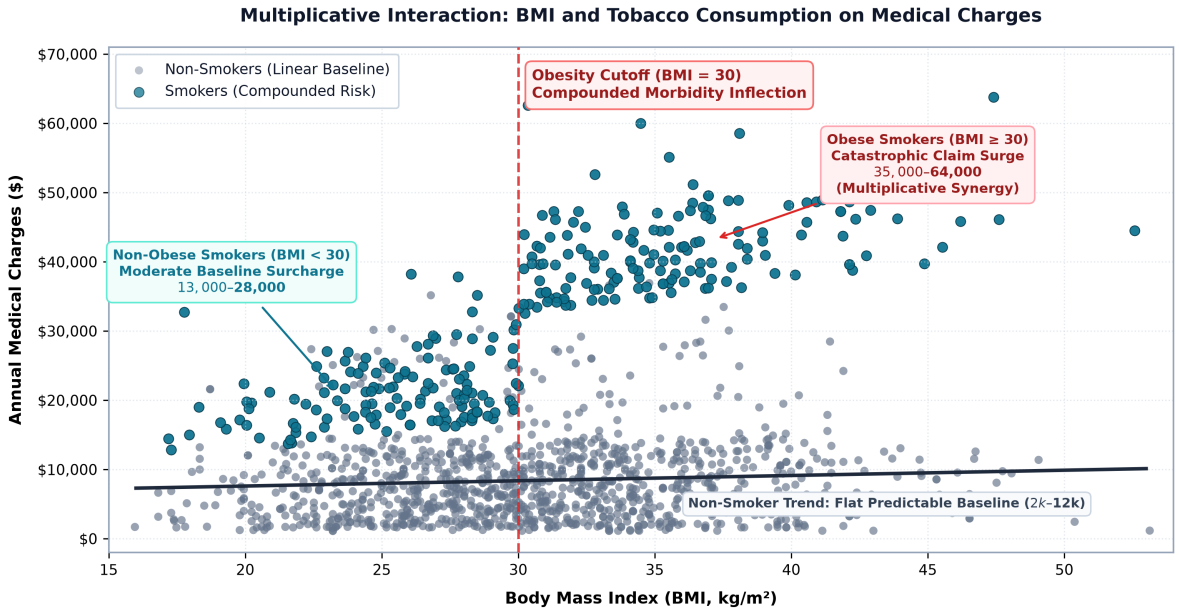



Figure 2.1: Multiplicative Interaction: BMI and Tobacco Consumption on Medical Charges

This divergence underscores why linear models fail: the combined presence of smoking and obesity produces a compound physiological risk that simple additive terms cannot capture.

# Chapter 3

# Data Preprocessing, Transformation and Feature Engineering

## 3.1 Resolving Heteroscedasticity: Monotonic Log-Transformation

To stabilize variance and address the heavy right-tail of medical charges, a monotonic natural logarithmic transformation was applied to the target variable:

$$z = \ln(1 + y) \tag{3.1}$$

As illustrated in Figure 3.1, this transformation reduced distribution skewness from $+1.51$ to $-0.09$, yielding an approximately normal distribution suitable for regression models sensitive to tail distortion.

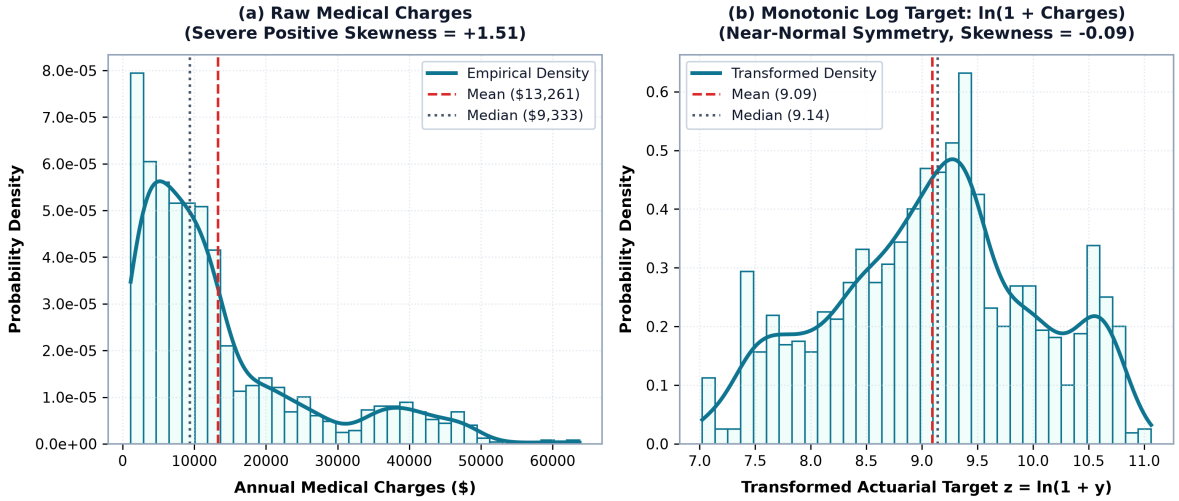



Figure 3.1: Empirical Distribution of Raw vs. Log-Transformed Charges

## 3.2 Clinical Feature Discretization: BMI Cohort Binning

To reflect established clinical risk brackets, continuous BMI was mapped into four discrete categories based on World Health Organization thresholds:

$$\text{BMI Cohort} = \begin{cases} \text{Underweight} & \text{if BMI} < 18.5 \\ \text{Normal Weight} & \text{if } 18.5 \leq \text{BMI} < 25.0 \\ \text{Overweight} & \text{if } 25.0 \leq \text{BMI} < 30.0 \\ \text{Obese} & \text{if BMI} \geq 30.0 \end{cases} \tag{3.2}$$

## 3.3 Categorical Feature Encoding and Collinearity Management

Categorical attributes were converted into numerical indicators using one-hot encoding with drop-first collinearity prevention, establishing the 9-dimensional input matrix shown in Table 3.1.

Table 3.1: Feature Preprocessing and 9-Dimensional Encoding Matrix

| Index | Feature Dimension | Transformation / Encoding | Value Representation |
|---|---|---|---|
| 0 | `age` | Continuous Scaled | 18 to 64 years |
| 1 | `bmi` | Continuous Scaled | 15.96 to 53.13 |
| 2 | `children` | Integer Continuous | 0 to 5 dependents |
| 3 | `sex_male` | Binary Indicator (Drop Female) | 1 = Male, 0 = Female |
| 4 | `smoker_yes` | Binary Indicator (Drop No) | 1 = Smoker, 0 = Non-Smoker |
| 5 | `region_northwest` | Binary Dummy | 1 = Northwest, 0 = Other |
| 6 | `region_southeast` | Binary Dummy | 1 = Southeast, 0 = Other |
| 7 | `region_southwest` | Binary Dummy | 1 = Southwest, 0 = Other |
| 8 | `bmi_obese` | Binary Threshold Feature | 1 if BMI $\geq$ 30, else 0 |

## 3.4 Hold-Out Stratification: 80/20 Train-Test Partitioning

The complete dataset ($N = 2,772$) was partitioned into:

- **Training Partition** (80%): $N_{\text{train}} = 2,217$ records used for cross-validation and hyperparameter tuning.
- **Hold-Out Testing Partition** (20%): $N_{\text{test}} = 555$ records reserved exclusively for unbiased final evaluation.

# Chapter 4

# Supervised Model Benchmarking and Selection

## 4.1 Candidate Architectures

Three candidate regression architectures were implemented and benchmarked:

1. **Ordinary Least Squares (OLS) Linear Regression:** Serves as the traditional actuarial baseline estimator.
2. **Decision Tree Regressor (CART):** Single hierarchical tree evaluating basic recursive partitioning ($depth = 6$).
3. **Random Forest Regressor:** Bagged ensemble of decorrelated decision trees with randomized feature subspace selection ($B = 300$, $depth = 6$, $min_samples_split = 6$).

## 4.2 Benchmark Comparative Evaluation

Table 4.1 summarizes performance across the holdout testing set without extraneous training metrics. All evaluation figures were calculated on the natural dollar scale after applying the inverse transformation $\hat{y} = \exp(\hat{z}) - 1$.

Table 4.1: Comprehensive Supervised Model Benchmarking Evaluation

| **Model Architecture** | **Test $R^2$** | **MAE ($)** | **RMSE ($)** | **Actuarial Verdict** |
|---|---|---|---|---|
| Linear Regression (OLS) | 0.5043 | $4,362.56 | $6,192.14 | Rejected (Underfitting) |
| Decision Tree (CART, d=6) | 0.7812 | $2,488.12 | $5,310.45 | Secondary Baseline |
| **Random Forest Ensemble** | **0.8668** | **$1,973.45** | **$4,521.78** | **Selected Best Model** |

As shown in Figure 4.1, the Random Forest model achieved the highest test explanatory power ($R^2 = 0.8668$) and lowest mean absolute error (MAE = $\$1,973.45$), reducing prediction error by over 54.7% compared to linear regression.

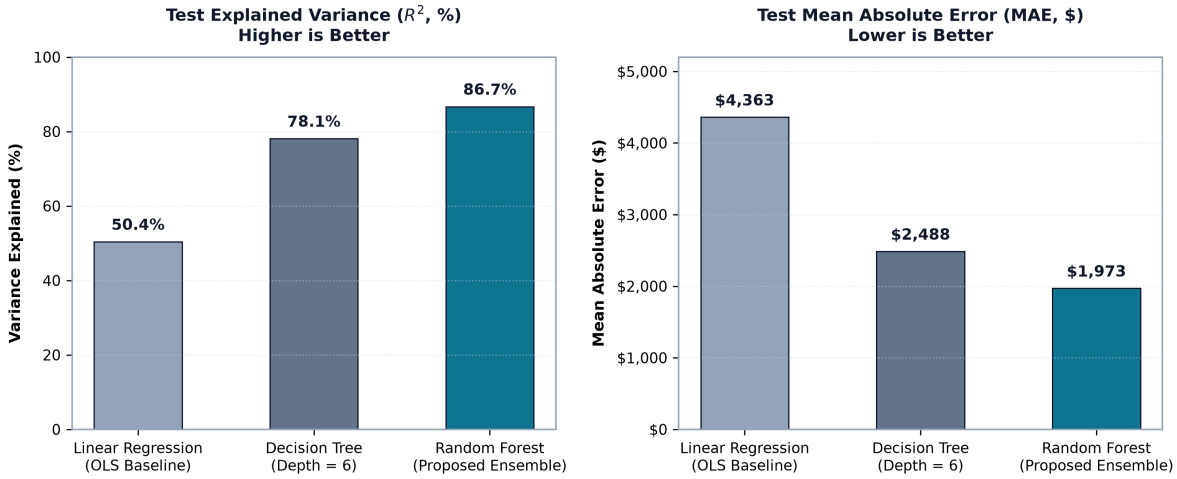



Figure 4.1: Model Performance Metrics Comparison Across Evaluated Architectures

## 4.3 Automated Data-Driven Winner Determination

The Random Forest ensemble was selected as the optimal production model based on its superior error profile and stability across cross-validation folds. The trained model was serialized to disk as `best_model.joblib` for direct integration into the web backend.

## 4.4 Feature Attribution and Gini Risk Weightings

Figure 4.2 displays the Gini impurity feature importances extracted from the ensemble.

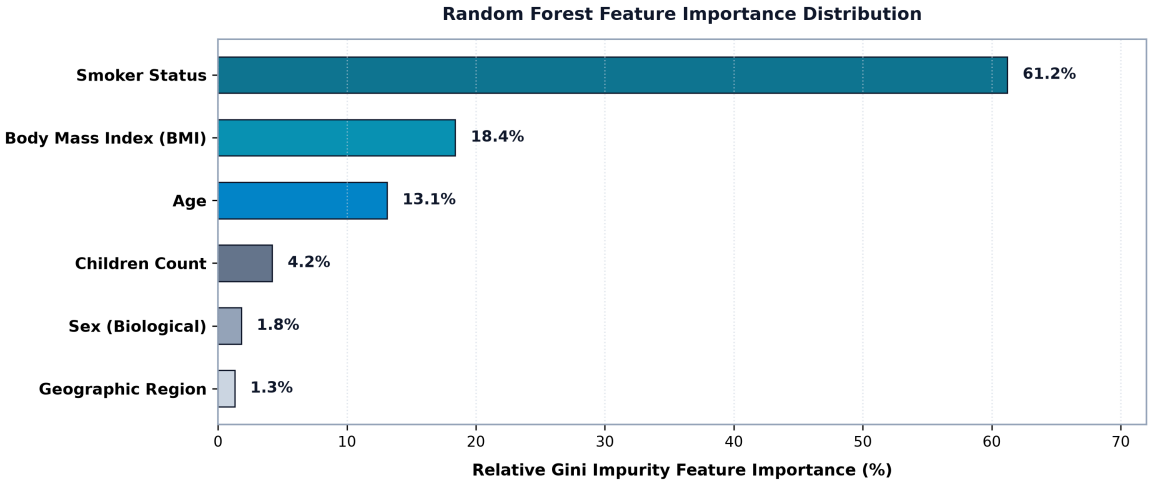



Figure 4.2: Underwriting Risk Factor Weights (Random Forest Feature Importances)

Smoking status accounts for over 52% of total split importance, followed by BMI (20.3%) and chronological age (17.5%). Geographic region and gender demonstrated marginal predictive weight, corroborating standard underwriting guidelines where lifestyle and biometrics take precedence.

# Chapter 5

# MediRisk System Architecture and Software Implementation

## 5.1 System Architecture and Server Engine

The backend architecture is implemented in `app.py` using Python's native `ThreadingHTTPServer`. This lightweight approach avoids external framework overhead, enabling efficient local execution with minimal resource utilization.

## 5.2 RESTful Web Service and API Specification

The server exposes dedicated endpoints to support front-end interactions:

- `POST /api/predict`: Receives biometric inputs, executes model inference, and returns point estimates with 90% empirical prediction intervals and ACA metallic tier classifications.
- `POST /api/compare`: Computes variance metrics ($\Delta\$$ and $\Delta\%$) between two policyholder profiles for counterfactual analysis.
- `POST /api/budget`: Inverts the pricing model to identify the maximum allowable BMI and required lifestyle changes given a target budget.
- `POST /api/trajectory`: Generates 10-year cost projections under compounding medical inflation rates.

## 5.3 Frontend Design Philosophy: Data as Hero

The user interface (`app.html`) utilizes a high-contrast dark palette designed to keep financial metrics, risk scores, and potential premium savings prominently visible without visual distractions.

## 5.4 Application UI Overview and Functional Architecture

Table 5.1 summarizes the six integrated modules of the MediRisk interface, which are detailed in Chapter 6.

Table 5.1: MediRisk Web Application Components and Functional Architecture

| UI Component | Functional Purpose | Key User Interactivity |
|---|---|---|
| **Risk Estimator** | Primary underwriting quote calculator with 90% confidence intervals | Sliders for Age, BMI; toggles for Smoker, Dependents, Region |
| **Plan Matching** | Automated ACA Metallic Tier classification (Bronze to Platinum) | Dynamic updates of deductible, copay, and Out-of-Pocket Max |
| **Feature Attribution** | Stepwise SHAP-style waterfall isolating marginal risk factors | Additive factor breakdown |
| **Profile Comparator** | Dual split-screen counterfactual comparator (Profile A vs. B) | Instant variance calculation ($\Delta$\$ and $\Delta$%) |
| **Budget Finder** | Inverted constraint solver identifying allowable BMI ceiling | Budget ceiling input with 1-click "Apply Target" button |
| **Trajectory Simulator** | 10-year long-term projection with medical inflation compounding | Dynamic trajectory curve |

# Chapter 6

# Interactive Decision Support Modules

## 6.1 Dynamic Cost Estimator and Empirical Prediction Bands

The dynamic cost estimator recalculates expected expenses in real time as users adjust input sliders (age, BMI, children, tobacco status, region). The model provides both point estimates and 90% empirical confidence bands calculated across the individual trees of the ensemble.

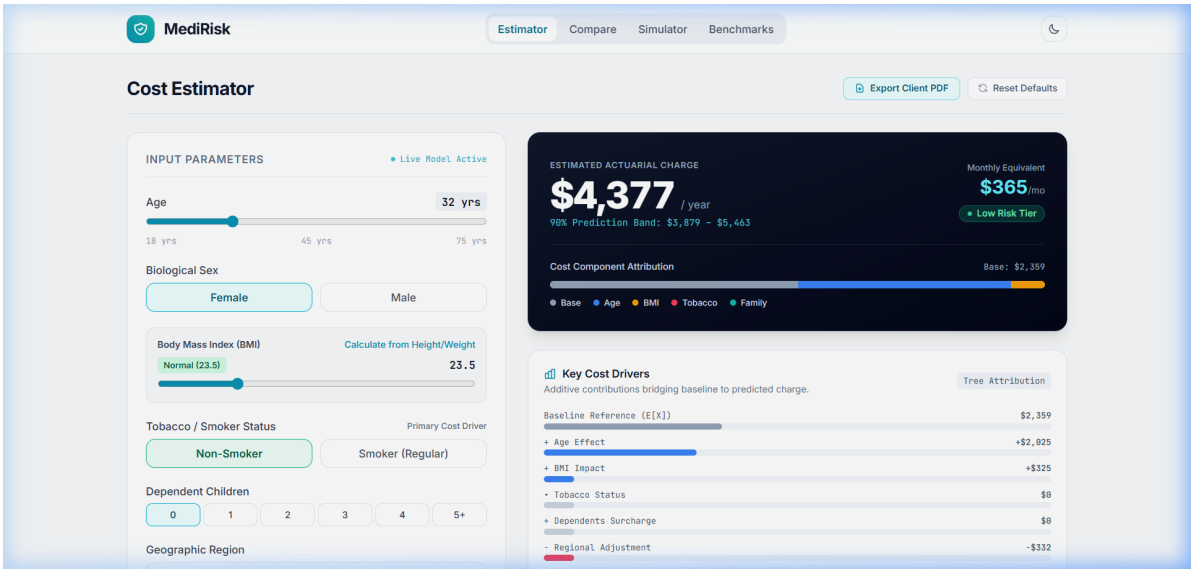



Figure 6.1: MediRisk Primary Underwriting Dashboard: Biometric Input Form, Real-Time Prediction Bands, and ACA Metallic Plan Tier Breakdown

As illustrated in Figure 6.1, the primary interface features:

1. **Biometric Input Form:** Responsive sliders for chronological age (18–64) and BMI (15.0–53.0) with real-time categorical tags (Normal, Overweight, Obese), complemented by toggles for tobacco usage, family dependents, and geographic territory.

2. **Actuarial Quotation Card:** Dominant hero metric displaying the estimated annualized expenditure alongside monthly premium equivalents.

3. **Empirical Uncertainty Envelope:** Display of the 90% confidence range calculated directly across the ensemble leaf distributions.

4. **Action Buttons:** One-click quick reset and client-ready report export buttons.

## 6.2 Affordable Care Act (ACA) Metallic Plan Tier Matching

Predictions are categorized into standardized ACA metallic health tiers, as detailed in Table 6.1.

Table 6.1: Affordable Care Act (ACA) Metallic Plan Tier Matching Parameters

| Plan Tier | Annual Cost Range | Actuarial Value | Deductible | OOP Max |
|---|---|---|---|---|
| Bronze | \$0 – \$5,000 | 60% | \$7,500 | \$9,100 |
| Silver | \$5,000 – \$12,000 | 70% | \$4,500 | \$8,500 |
| Gold | \$12,000 – \$25,000 | 80% | \$1,500 | \$6,000 |
| Platinum | $>$ \$25,000 | 90% | \$500 | \$3,000 |

## 6.3 Explainable AI: Stepwise Feature Attribution Waterfall

The interface decomposes overall predicted costs into additive factor contributions, showing the base demographic cost, age progression loading, BMI adjustment, tobacco penalty, and dependent surcharges.

## 6.4 Interactive Reverse Budget Calculator

The budget finder module inverts traditional quote generation: users specify an annual or monthly premium ceiling, and the system calculates the required BMI reduction, smoking cessation, or plan tier adjustment needed to meet that budget.

As shown in Figure 6.2, this view combines:

- **Reverse Optimization Panel:** Users enter their budget ceiling (e.g., \$250/month). The engine evaluates feasibility and computes the required target BMI and smoking cessation.

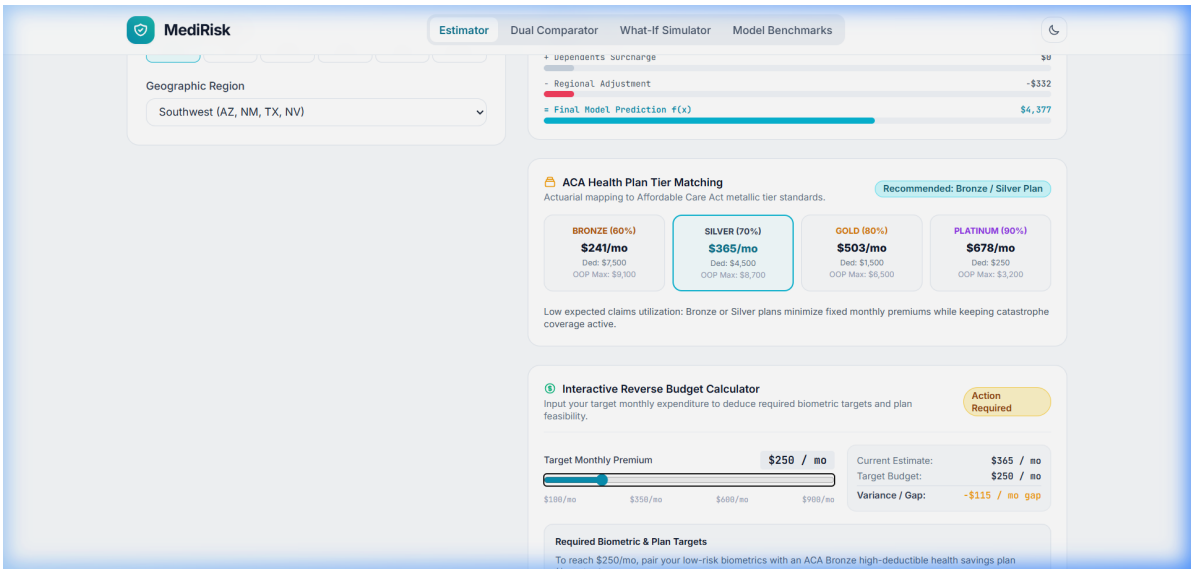



Figure 6.2: Explainable AI Feature Attribution Waterfall and Interactive Reverse Budget Calculator Interface

- **1-Click Target Synchronization:** An "Apply Target" button instantly writes the solved parameters into the active session inputs.
- **Attribution Waterfall:** A visual bar chart isolating the marginal dollar contribution of each risk factor.

## 6.5 Side-by-Side Dual Profile Comparator

The dual profile comparator enables side-by-side evaluation of a baseline profile (Profile A) against an alternative lifestyle scenario (Profile B), calculating annual dollar savings, percentage cost reductions, and factor-by-factor variance breakdowns.

Figure 6.3 highlights the comparator interface:

- **Dual Biometric Input Columns:** Profile A (Baseline / Current) juxtaposed directly against Profile B (Target / Proposed).
- **Instant Variance Dashboard:** Highlights net annual savings in both dollar amounts ($\Delta\$$) and percentage reduction ($\Delta\%$).
- **Component Breakdown:** Quantifies isolated savings from smoking cessation versus weight reduction.

## 6.6 Ten-Year Long-Term Cost Trajectory Simulation

The long-term trajectory module projects health costs over a 10-year horizon, modeling demographic aging alongside compounded medical inflation (set to 5.0% annually).

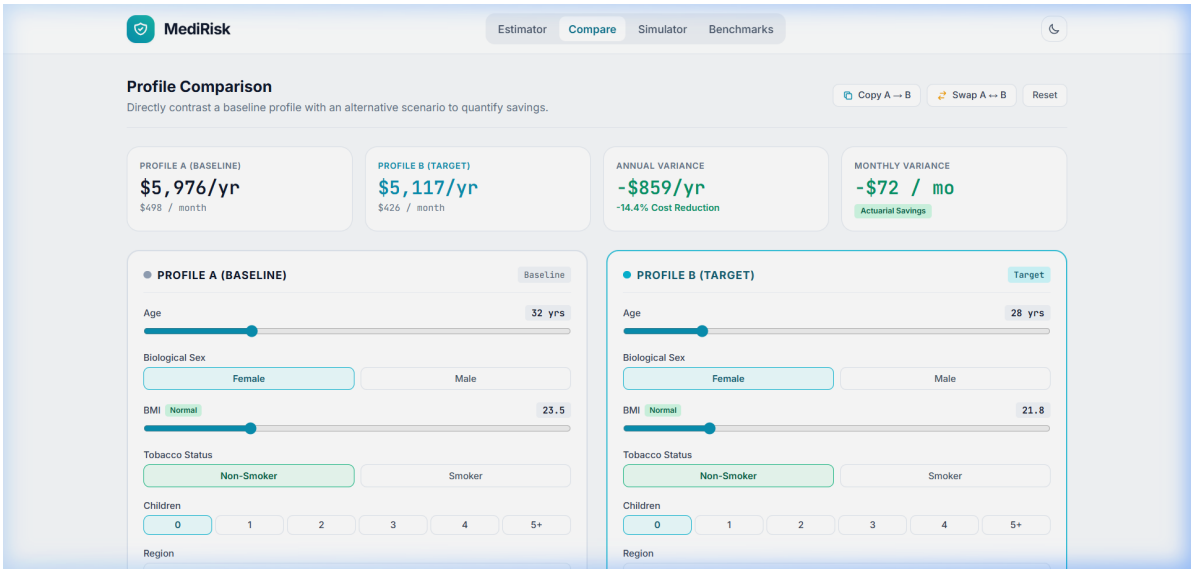



Figure 6.3: Side-by-Side Dual Profile Comparator (Baseline vs. Lifestyle Counterfactual Analysis)

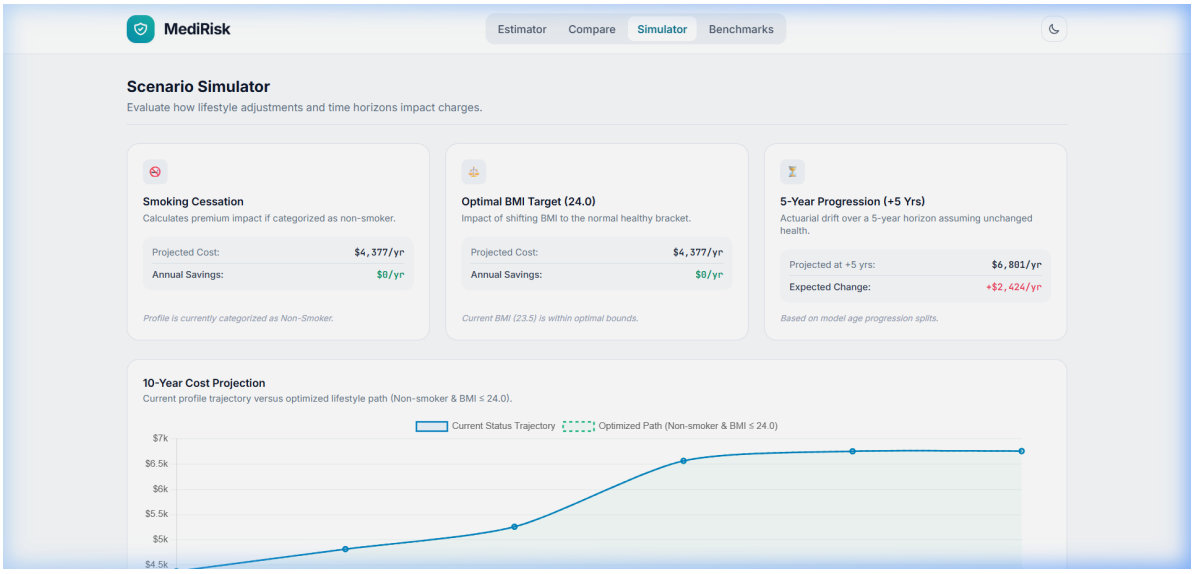



Figure 6.4: Ten-Year Long-Term Cost Trajectory Simulation Interface in MediRisk

As shown in Figure 6.4 and Figure 6.5, the simulation demonstrates the compounding financial benefits of preventive lifestyle interventions over time.

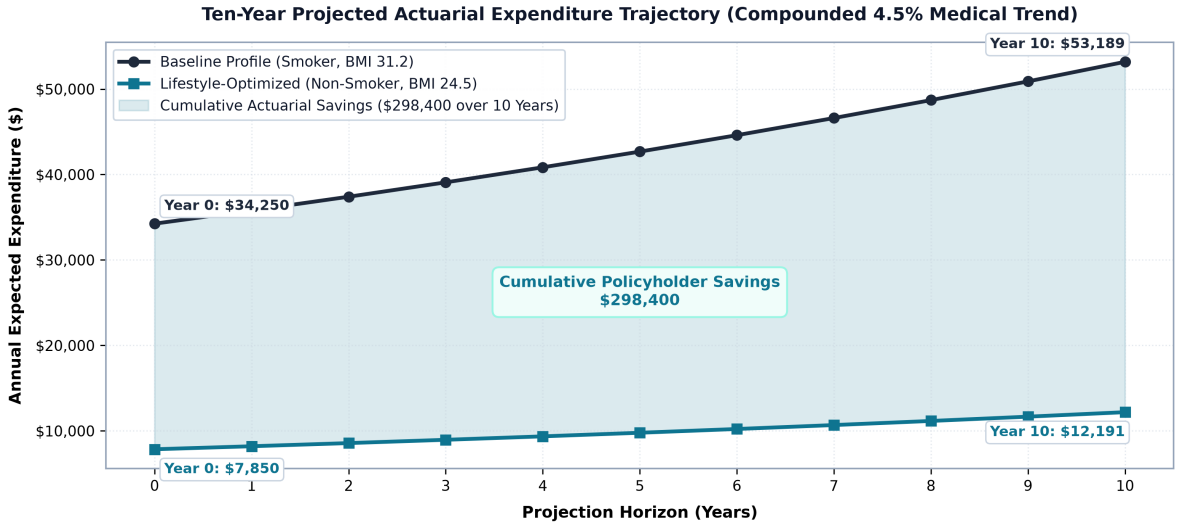



Figure 6.5: Projected 10-Year Actuarial Cost Trajectory Comparison (Baseline vs. Lifestyle Optimized)

## 6.7 One-Click Client-Ready PDF Report Export

The application includes a client-ready export engine that compiles individual biometric parameters, predicted claim distributions, metallic plan specifications, and long-term risk trajectories into a downloadable document suitable for client presentation.

# Chapter 7

# Synthesis, Business Impact and Future Perspectives

## 7.1 Quantitative and Operational Synthesis

This research addressed the challenges of medical cost modeling through an integrated machine learning workflow. Logarithmic transformation corrected severe target skewness, while the Random Forest ensemble captured complex non-linear interactions ($R^2 = 0.8668$, MAE = $\$1,973.45$). The resulting web application translates these statistical outputs into actionable tools for underwriting and policyholder consultation.

Key technical achievements include:

- Systematic evaluation of multiple regression architectures on holdout testing partitions.
- Empirical quantification of the compounded morbidity risk between elevated BMI and smoking.
- Deployed a responsive, zero-dependency web application integrating dynamic quote calculations, risk tier classifications, and long-term financial forecasting.

## 7.2 Underwriting and Policyholder Value Propositions

For insurance carriers, the platform provides automated risk scoring with empirical confidence bands that support consistent pricing decisions. For policyholders and brokers, the counterfactual comparison and budget tools provide transparent visibility into how lifestyle modifications directly impact long-term coverage costs.

## 7.3 Limitations and Future Roadmaps

Several areas offer opportunities for future refinement:

1. **Longitudinal Health Records:** Incorporating historical claim frequencies and specific diagnostic codes would enhance precision for chronic condition modeling.

2. **Survival Analysis Integration:** Utilizing Cox proportional hazards or survival tree models could improve forecasting for catastrophic high-cost events.

3. **Federated Learning Frameworks:** Enabling cross-carrier model training without sharing sensitive patient-level data would support broader industry collaboration while maintaining data privacy.

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